

Smart Water Pouring System

IOT102-SE1822, Group 1

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Abstract

This project presents the design and implementation of a Smart Water Pouring System that leverages IoT technologies to optimize water usage in irrigation applications. The system is built around an Arduino Uno as the central controller with a manual control interface using RemoteXY. The system features three preset buttons—'Low' (50 ml), 'Medium' (100 ml), and 'High' (200 ml)—as well as an input field for specifying custom water volumes in milliliters. A water level sensor is incorporated to ensure that water is dispensed only when an adequate amount is available in the reservoir, while a relay module controls the water pump. Additionally, the system integrates a Bluetooth module with RemoteXY and an I2C LCD display for real-time local monitoring and remote control via a mobile application. The methodology involves calibrating the water level sensor to accurately assess water availability and enable decision-making for water pump activation. Experimental results demonstrate that the system can effectively dispense precise volumes of water while minimizing wastage. This work not only provides a practical solution for smart irrigation but also offers an educational platform for students to gain hands-on experience in designing, programming, and integrating IoT devices. Future enhancements may include cloud connectivity for data logging and advanced analytics to further improve water management strategies.

I. INTRODUCTION

The Internet of Things (IoT) represents a transformative paradigm in modern technology, where everyday objects are embedded with sensors, processors, and communication interfaces to connect and exchange data over networks. This interconnectivity allows devices—from household appliances to industrial machines—to be monitored and controlled remotely, leading to smarter and more efficient operations. The benefits of IoT are far-reaching; it enables real-time data collection and analysis, improves decision-making through predictive insights, and optimizes resource management across various sectors, including agriculture, healthcare, manufacturing, and smart cities [1]. Furthermore, IoT systems facilitate decentralized data processing, often performing computations at the edge to reduce latency and enhance responsiveness, which is crucial for time-sensitive applications [2] [3]. The main components of IoT are shown in figure 1.

In academic projects, integrating IoT technologies not only exposes students to cutting-edge tools but also enhances their creative problem-solving and real-world engineering skills [4]. The "Smart Water Pouring System" project is a prime example, designed to optimize water usage in irrigation through direct and remote control. Traditionally, manual water pumping methods have been widely used in small-scale irrigation, but they come with significant limitations. These include inconsistent water distribution due to human error, excessive water wastage from over-pouring, and the labor-intensive nature of the process, which becomes impractical for larger areas or frequent irrigation needs. Moreover, manual systems lack real-time feedback, making it difficult to adjust water volumes based on immediate environmental conditions or reservoir levels, often leading to inefficient resource use and potential crop damage from under- or over-watering.

The "Smart Water Pouring System" addresses these shortcomings by introducing a highly functional and automated alternative. The system utilizes three dedicated buttons—labeled as "Low" (50 ml), "Medium" (100 ml), and "High" (200 ml)—along with an input field for custom water volumes, allowing users to precisely control the amount of water dispensed with minimal effort. This precision eliminates the guesswork and inconsistency inherent in manual pouring, ensuring that plants receive the exact amount of water required. The core of the system is built around an Arduino Uno, which integrates a Bluetooth module with RemoteXY for remote control via a mobile application and an I2C LCD display for real-time local monitoring. This dual-control capability—local and remote—offers flexibility, enabling users to manage irrigation from a distance, a significant improvement over manual methods that require physical presence. Additionally, a water level sensor is employed to verify that sufficient water is available in the reservoir before the relay-controlled pump is activated, preventing dry runs that could damage the pump or disrupt irrigation—a common issue in traditional setups where water levels are checked manually or overlooked entirely.

By calibrating the water level sensor and fine-tuning the manual control interface, the system achieves accurate water dispensing and minimizes wastage, directly tackling the inefficiencies of manual pumping. Experimental results have demonstrated that this approach is both reliable and efficient, providing a valuable educational platform for students to gain hands-on experience in IoT device design, programming, and real-time data analysis. The system's improvements over manual methods are evident in its ability to automate repetitive tasks, reduce physical labor, and provide data-driven control, all of which contribute to sustainable water use. Future enhancements may include cloud connectivity for extensive data logging and advanced analytics

to further refine water management strategies. For instance, integrating weather data or soil moisture sensors could enable the system to dynamically adjust water volumes based on real-time environmental conditions, taking automation and efficiency to the next level. Such advancements would not only enhance the system’s utility in educational settings but also make it a practical tool for small-scale farmers seeking to modernize irrigation practices while conserving resources.

At the heart of the system is an Arduino Uno, which integrates a Bluetooth module with RemoteXY for remote control via a mobile application, as well as an I2C LCD display for real-time local monitoring. A water level sensor is incorporated to ensure that the water reservoir contains a sufficient supply before the pump, controlled by a relay, is activated. By addressing the limitations of manual water pumping—such as labor demands, inaccuracy, and lack of feedback—this design not only enhances operational efficiency but also provides a hands-on educational platform for students, enabling them to gain practical experience in IoT device design, programming, and real-time data analysis [4]. Future enhancements may involve integrating cloud connectivity for extensive data logging and advanced analytics, further refining water management strategies and contributing to sustainable agricultural practices.



Fig. 1. Major components of IoT system.

II. METHODS AND MATERIALS

A. System Model and Block Diagram

The developed system comprises both hardware and software components meticulously designed to optimize water management through IoT technologies, as illustrated in the block diagram in figure 2. On the hardware side, the system is centered around an Arduino Uno R3, which acts as the primary microcontroller for local data processing and control decisions. The system integrates a water level sensor (YF-S401), which ensures the reservoir contains sufficient water before the pump is activated, preventing dry runs that could damage the pump or disrupt irrigation. Additionally, a water flow sensor (also YF-S401, as indicated in the diagram) accurately measures the volume of water dispensed, providing precise feedback to the system for better resource management. These sensors feed real-time data into the Arduino Uno, enabling it to make informed decisions about water dispensing operations.

For user interaction, the system features a manual control interface consisting of three push buttons labeled “Low” (50 ml), “Medium” (100 ml), and “High” (200 ml), as shown in the block diagram. These buttons allow users to select predefined water volumes, while an additional input field (accessible via the mobile application) enables the specification of custom water volumes, offering flexibility for diverse irrigation needs. This design ensures that users can trigger water dispensing operations on demand, independent of automatic sensor-based measurements, while still benefiting from the system’s precision and automation capabilities. A relay module is employed to control the water pump based on sensor readings and manual inputs, ensuring that water is dispensed only when necessary, thereby conserving resources and minimizing wastage.

For real-time feedback, the system integrates an I2C LCD display, which provides vital information such as water level status, dispensed water volume, and system status updates, enhancing local monitoring capabilities. In parallel, a Bluetooth module facilitates wireless connectivity, enabling seamless communication with smart devices such as smartphones, as depicted in the “Connectivity & Analytics” section of the block diagram. Through this Bluetooth connection, users can remotely monitor and control the system via a mobile application (e.g., using the RemoteXY platform mentioned earlier), visualizing real-time data such as water levels and flow rates. This remote access not only improves user convenience but also allows data to be uploaded to an IoT platform for further logging and analysis, supporting long-term water usage optimization.

On the software side, firmware developed in the Arduino IDE orchestrates the system’s operations, including sensor readings, relay control, LCD updates, and Bluetooth communications. These routines manage thresholds for water levels, process user-triggered actions from the push buttons or mobile app, and ensure efficient water usage by coordinating the pump’s operation with sensor data. The block diagram highlights the clear division between the hardware and software components (“Application

Field: Hardware & Software”) and the connectivity layer (“Connectivity & Analytics”), illustrating how the system bridges local control with remote monitoring.

To further enhance the system, future improvements could include the integration of additional sensors, such as soil moisture sensors, to enable dynamic water dispensing based on real-time environmental conditions. Additionally, incorporating cloud connectivity could allow for extensive data logging and advanced analytics, such as predictive water usage models or integration with weather forecasts to optimize irrigation schedules. Such enhancements would build on the existing Bluetooth-based remote control, making the system more adaptive and scalable for larger agricultural applications while maintaining its educational value for students learning IoT system design.

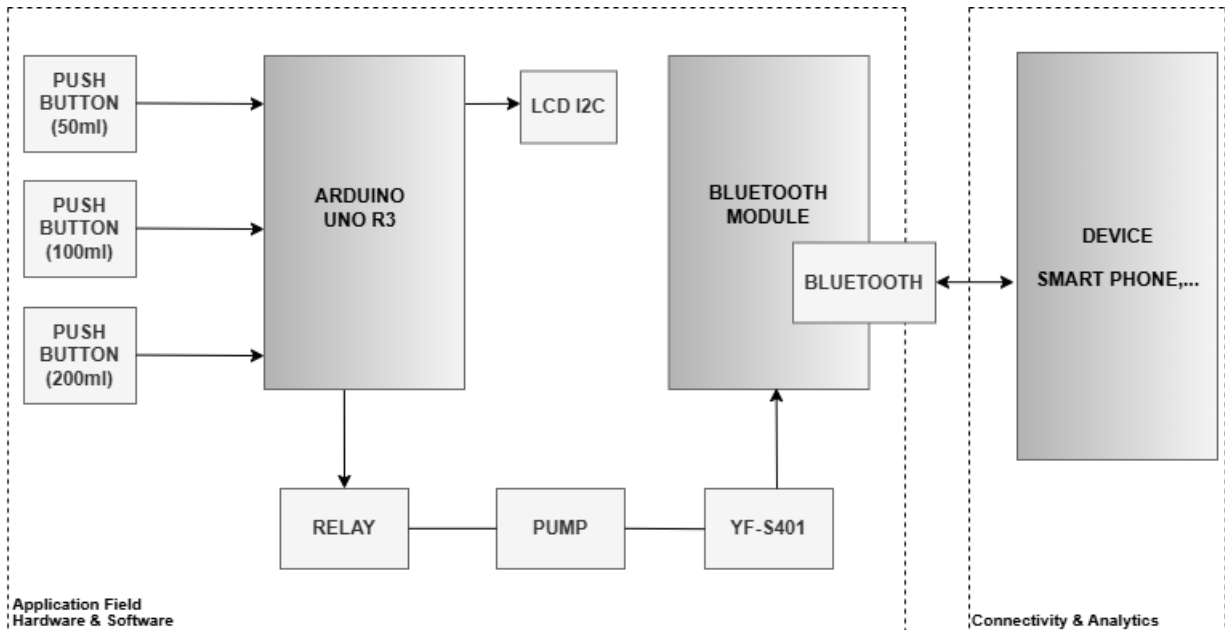


Fig. 2. Block diagram of the developed system.

B. Components and Peripheral Devices

- **Arduino Uno:** Arduino UNO is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.

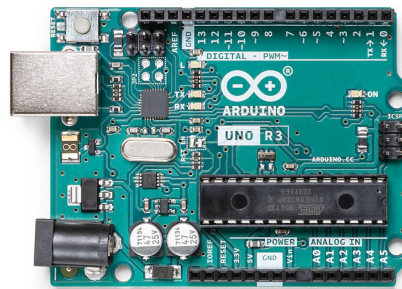


Fig. 3. Arduino Uno Rev3.

- **Bluetooth module:** The HC-05 is a Bluetooth module based on the CSR BC417 chip. It has 6 pins: VCC, GND, TXD, RXD, STATE, and EN (Key). It operates at 3.3V logic level but can be powered by 5V. The module supports both master and slave modes and communicates via serial UART. It can be used to wirelessly transmit data between microcontrollers, computers, or smartphones. To use it, simply connect it to the Arduino using the TX and RX pins and pair it with a Bluetooth device to start sending and receiving data.



Fig. 4. Bluetooth module HC-05.

- **Relay module:** 5V relay module is a control board that turns on/off the electrical circuit by signals from a microcontroller. It has 3 input pins: VCC, GND and IN. When receiving a control signal from a microcontroller (e.g. Arduino), the relay will turn on or off the high voltage circuit (AC or DC). The module has an LED to display the operating status. Just supply 5V power, connect the control signal and you can easily control electrical devices remotely.



Fig. 5. Relay module.

- **Water Pump:** Submersible DC Water Pump is a compact device designed to move water in small-scale applications, such as aquariums, fountains, or hydroponic systems. It typically operates on a DC voltage range (for example, 3–12V), with two wires for power input (red for positive and black for negative). The pump has an inlet and an outlet for water flow, along with an internal impeller that circulates water when powered. Its sealed housing allows it to be fully submerged, and its lightweight design makes it easy to integrate into a variety of projects. It contains everything needed to function as a simple water pump; simply connect it to a suitable DC power source or battery to get started.



Fig. 6. Water Pump.

- **YS-S401:** A water flow sensor is a device that measures the speed of water through a pipe. It has three connecting wires: red (VCC) for power supply, black (GND) for ground, and yellow (signal) for transmitting pulse data. When water flows through, the internal impeller rotates and generates a pulse signal, which helps the microcontroller calculate the water flow. Just supply power and connect the signal wire to the microcontroller, you can easily measure the amount of water passing through the system.



Fig. 7. YS-S401.

Figure 8 shows the circuit interfacing schematic of the developed system. The water level sensor is connected to pin A1 of the Arduino. The relay, which is responsible for controlling the water dispenser, is connected to pin 8. The LCD display is connected to the Arduino via pins A4 (SDA) and A5 (SCL) using I2C communication. The Bluetooth module (RemoteXY) is connected to the Arduino via pins 10 (RX) and 11 (TX), allowing wireless communication and control.

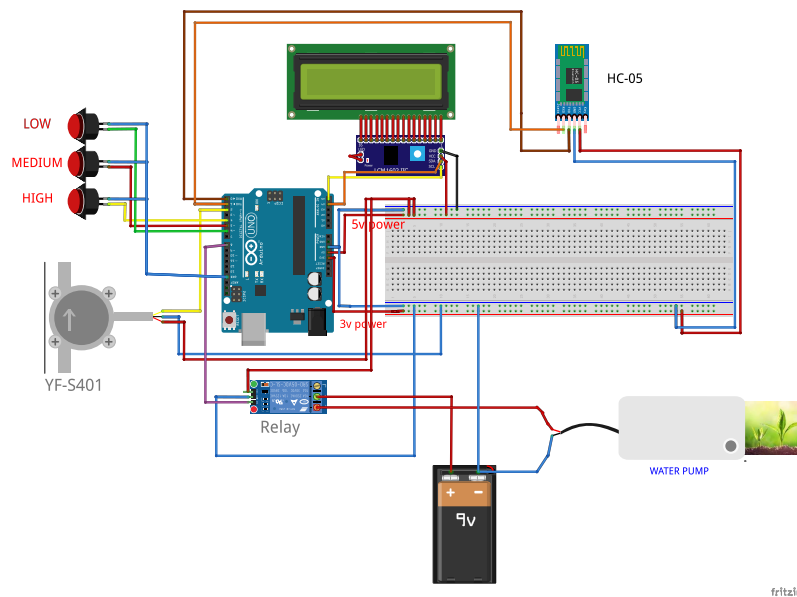


Fig. 8. Circuit schematic/hardware interfacing of the proposed system.

The water level balancer system uses IoT technologies to manage water precisely, integrating components like an Arduino Uno R3 (ATmega328P-based), a Bluetooth module (HC-05) for remote control via pins 10 (TX) and 11 (RX), a water level sensor, and a YF-S401 water flow sensor (connected to pin 2) to monitor water parameters. A 5V relay (pin 8) controls a mini submersible pump, preventing dry running, while three push buttons (A4, A5, A6) allow manual selection of water volumes (50 ml, 100 ml, 200 ml). A 16x2 LCD (I2C, pins A4 SDA, A5 SCL) provides real-time feedback on water level and flow. The system, prototyped using a breadboard (MB-102) and jumper wires, is powered via Arduino's 5V and GND pins. Arduino firmware manages sensor data, user inputs, relay control, and LCD updates. Tables I and II detail the pin connections and component list, respectively. Future enhancements could include a Wi-Fi module for cloud connectivity and a soil moisture sensor for dynamic water adjustment, enhancing scalability and sustainability for educational and irrigation applications.

TABLE I
INTERFACING BETWEEN ARDUINO UNO AND ITS COMPONENTS (PIN-TO-PIN)

Arduino Uno	Water Level Sensor	LCD 12C	Relay	Bluetooth Module (HC-05)	Water Flow Sensor (YF-S401)
GND	GND	GND	GND	GND	GND
5V	VCC	VCC		VCC	VCC
A4	LOW (Button)				
A5	MEDIUM (Button)				
A6	HIGH (Button)				
2					Data (Pulse Signal)
8			IN		
10				TX	
11				RX	
A4		SDA			
A5		SCL			

TABLE II
SYSTEM'S COMPONENTS AND PERIPHERAL DEVICES

Components/devices	ID/Remarks
Arduino Uno R3	ATmega328P based
Bluetooth Module	HC-05
Water Flow Meter	YF-S401, hall-effect
Push Button	3 tactile (LOW, MEDIUM, HIGH)
Liquid-Crystal Display (LCD)	16 × 2 LCD with I2C module
Relay	5V relay module
Pump	Mini submersible
Breadboard	MB-102, full
Connecting Wires	Jumper, MM MF FF

C. Software Programming

Bluetooth is a widely adopted wireless communication technology that plays a pivotal role in numerous IoT applications, offering a reliable and efficient means of connecting devices in close proximity. Its significant advantages include low power consumption, cost-effectiveness, robust security, and the ability to operate without reliance on a cloud server, making it an ideal choice for localized device communication [5]. In our project, the smart water pouring system leverages Bluetooth to establish a secure and direct connection between the Arduino-based control system (using an Arduino Uno R3) and the RemoteXY interface running on a mobile device. Specifically, the HC-05 Bluetooth module, connected to the Arduino via pins 10 (TX) and 11 (RX) for serial communication, facilitates this interaction. This setup enables real-time data exchange—such as water level and flow rate updates—and remote control of the system, allowing users to monitor and adjust water dispensing parameters directly from their smartphones. By avoiding dependence on external servers, the Bluetooth-based architecture minimizes latency, simplifies the system design, and reduces potential points of failure, such as internet connectivity issues. The localized communication provided by Bluetooth ensures that commands (e.g., activating the pump for a specific water volume) and monitoring data are transmitted efficiently and securely, enhancing the overall responsiveness and reliability of the smart water pouring system. However, Bluetooth's range limitation (typically up to 10 meters for the HC-05 module) may restrict its use in larger-scale applications, a challenge that could be addressed in future iterations by integrating alternative technologies like Wi-Fi for extended coverage.

To be recognized as part of the IoT ecosystem, end-point devices must communicate efficiently with data, a requirement that our system fulfills through a Bluetooth-based solution integrated with RemoteXY. This integration involves several key steps: establishing a secure Bluetooth connection between the HC-05 module and the mobile device, configuring RemoteXY as the user interface for control (e.g., buttons for selecting water volumes) and data visualization (e.g., displaying water level status), and embedding the necessary communication protocols within the Arduino firmware. The firmware, developed in the Arduino IDE, ensures seamless data exchange by handling Bluetooth serial communication and processing user commands in real time. Bluetooth offers several key benefits for IoT applications, particularly in the context of this project. For instance, Bluetooth Low Energy (BLE)—while not used in the HC-05 module but available in newer modules like the HM-10—significantly reduces power consumption, which is crucial for battery-operated devices [6]. Moreover, Bluetooth is cost-effective, widely compatible with a broad range of consumer electronics (e.g., smartphones, tablets), and supports secure localized data exchange, thereby reducing latency and enhancing system responsiveness. In addition, its ease of deployment and scalability facilitate the setup of rapid, flexible, and robust IoT systems, making it a practical choice for educational projects and small-scale applications like ours [7]. Despite these advantages, Bluetooth's limited bandwidth and range may pose challenges for applications requiring high data

throughput or long-distance communication. To address this, future enhancements could involve integrating a dual-connectivity approach, combining Bluetooth for local control with a Wi-Fi module (e.g., ESP8266) for cloud connectivity, enabling remote data logging and advanced analytics while retaining the benefits of Bluetooth's low-power, localized communication.

Steps to Create the RemoteXY Interface: To enable remote control and monitoring of the smart water pouring system via Bluetooth, the RemoteXY platform is used to create a user-friendly mobile interface. Below are the steps to design and implement the RemoteXY interface for this project:

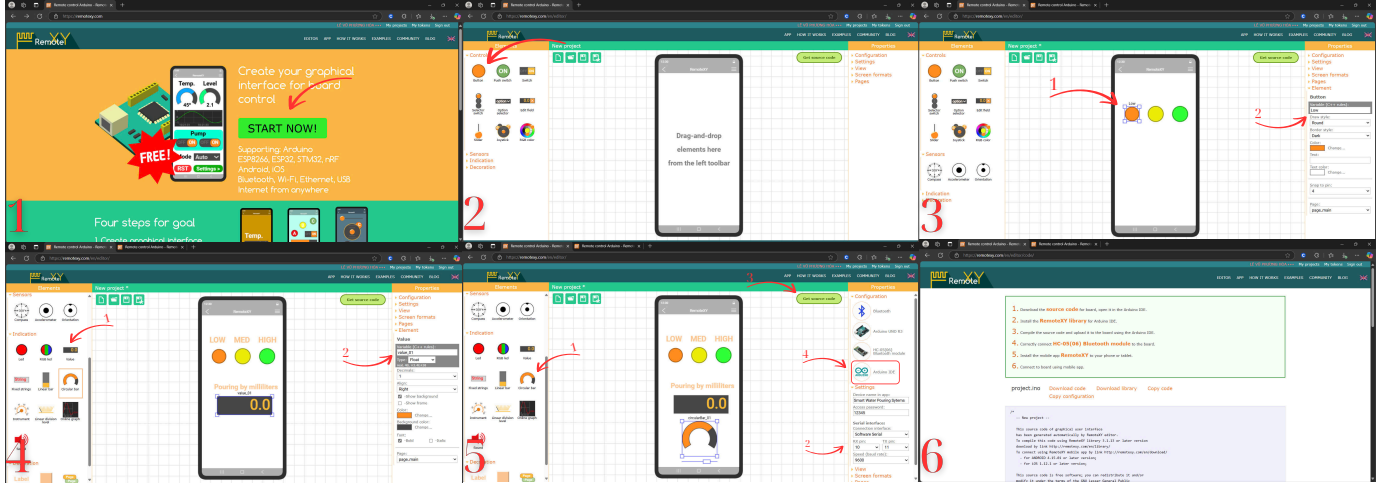


Fig. 9. Steps to Create the RemoteXY Interface.

The water pouring system operates via a RemoteXY interface that enables manual control of the water pump. The interface includes three preset buttons—'Low' (50 ml), 'Medium' (100 ml), and 'High' (200 ml)—as well as an input field that allows users to specify a custom volume in milliliters. When a button is pressed or a custom value is entered, the system activates a relay connected to the water pump for a duration proportional to the desired volume (assuming 10 ms per ml) as shown in figure 9. An attached LCD displays messages indicating the amount of water being poured and confirms when the process is complete. This design focuses solely on manual water control without relying on sensor-based automation.

```
if (!pumpOn && RemoteXY.connect_flag == 1) {
  if (RemoteXY.button_small) {
    startPump(50);
  } else if (RemoteXY.button_medium) {
    startPump(100);
  } else if (RemoteXY.button_large) {
    startPump(200);
  } else if (RemoteXY.input_ml > 0) {
    startPump(RemoteXY.input_ml);
    RemoteXY.input_ml = 0; // Reset input
  }
}
```

Water Flow Meter: In this system, when the pump control is turned on, water flows through the YS-S401 sensor. This water flow meter is equipped with an integrated magnetic hall-effect sensor that generates an electrical pulse with every complete rotation of its turbine. By counting these pulses, the system calculates the water flow rate. Although the YF-S401 is one of the most cost-effective water flow sensors available, its accuracy depends heavily on proper installation and calibration. According to its specifications, the sensor outputs a pulse frequency proportional to the water flow rate according to the formula:

$$F = 98 \times Q (\pm 2\% \text{ error})$$

where F is the frequency in Hertz and Q is the flow rate in liters per minute. This means that for a flow rate of 1 L/min, the sensor produces about 98 pulses per minute, implying that each pulse corresponds to approximately 1 / 98 L, or roughly 10.2 ml, of water. From the previous equation of water flow rate, the volume of water can be calculated as follows:

$$\text{FlowRate} : Q = \frac{F}{98}$$

$$WaterVolume = \frac{Q \times t(s)}{60}$$

where t is the time elapsed for water flow in seconds:

$$t(s) = t(current) - t(start)$$

D. Programming Flowchart

According to the diagram, each time a bluetooth button signal or physical button is received, the system will check the amount of available water through the water level indicator and decide whether to turn the water pump on or off. The system will record the water flow rate and volume using the water flow meter and send the data to RemoteXY. The diagram is shown in figure 10.

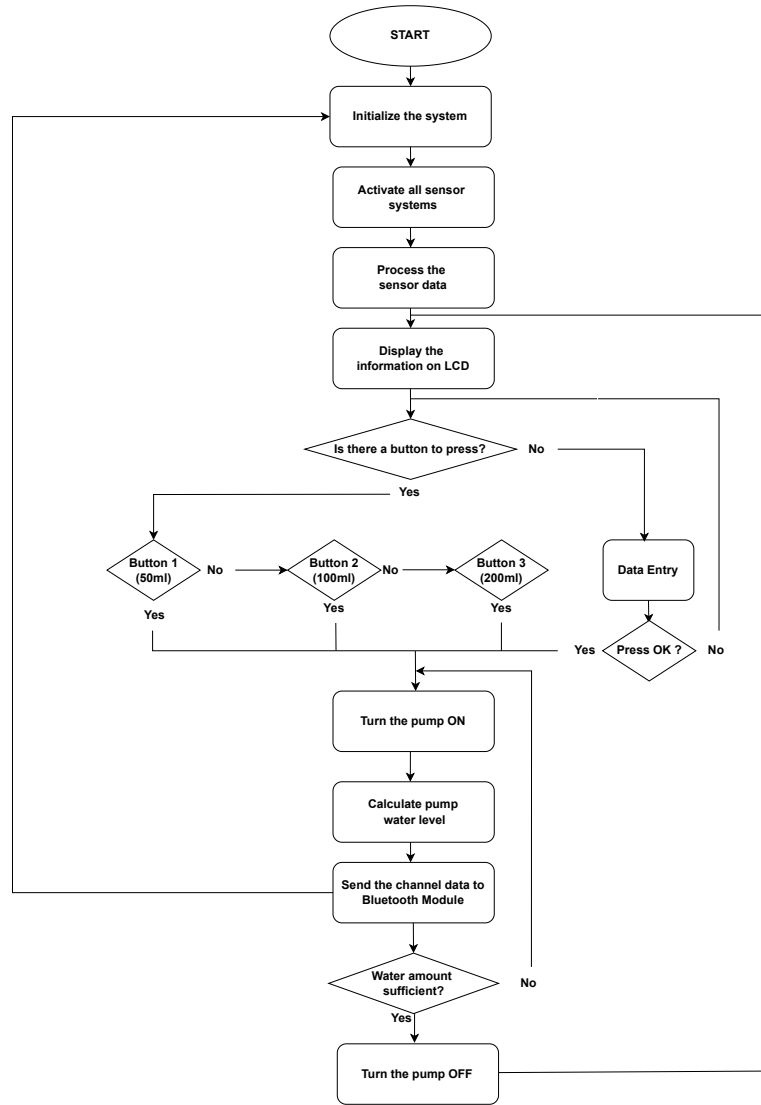


Fig. 10. Programming flowchart of the developed system.

III. RESULTS AND DISCUSSION

A. Prototype Implementation

Process of Connecting RemoteXY: To enable remote control and monitoring of the Smart Water Pouring System via the RemoteXY interface, a Bluetooth connection must be established between the HC-05 module and the mobile device running the RemoteXY app. The following steps outline the process of connecting RemoteXY to the system:

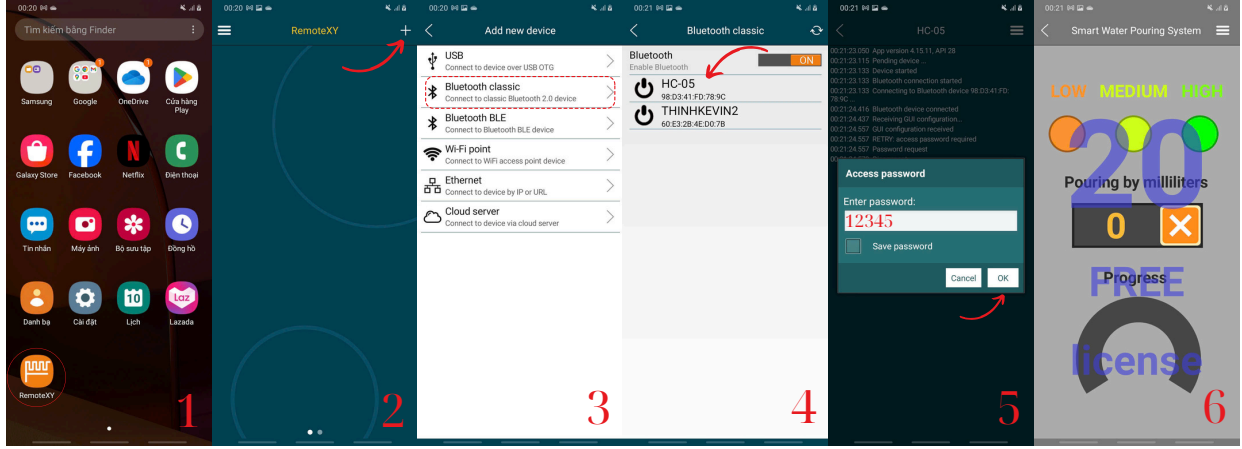


Fig. 11. How to connect with Remote XY.

System Design and Functionality: The Smart Water Pouring System is designed to provide an advanced automation solution for water pouring control, with the Arduino Uno R3 (ATmega328P-based) acting as the central controller. The Arduino's primary tasks in this system include receiving commands from the user (via a mobile app or physical buttons), processing data from sensors, and controlling devices such as the water pump. A key highlight of the system is the HC-05 Bluetooth module, which is directly connected to the Arduino Uno via pins 10 (TX) and 11 (RX) to establish wireless communication between the system and a smartphone application. The HC-05 module acts as a bridge, enabling users to send commands remotely and monitor system parameters without physically interacting with the hardware, thus enhancing user convenience and accessibility, especially in scenarios where direct access to the system is impractical.

The smartphone application, developed using the RemoteXY platform as described earlier, features a user-friendly interface tailored for the smart water pouring system. The interface includes three buttons labeled "Low" (50 ml), "Medium" (100 ml), and "High" (200 ml), aligning with the predefined water volumes specified in the system design, allowing users to select the desired amount of water to dispense. This remote control capability via Bluetooth provides significant convenience, particularly in situations where users are not in close proximity to the device, such as when managing irrigation in a garden or greenhouse from indoors. In addition to wireless control, the system is equipped with three physical push buttons mounted on the breadboard (MB-102), connected to analog pins A4, A5, and A6 of the Arduino, as detailed in Table I. These buttons are programmed to replicate the functions of the "Low," "Medium," and "High" modes in the app, enabling users to select the desired water volume (50 ml, 100 ml, or 200 ml) without needing their smartphone. This feature is particularly useful in emergency situations—such as when the phone is unavailable or the Bluetooth connection fails—or for users who prefer a quick, hands-on approach without opening the app. By integrating both wireless and physical control methods, the system offers increased flexibility and ensures accessibility for a wide range of users, from tech-savvy individuals to those who prioritize simplicity.

The YF-S401 water flow sensor, a core component of the system, is connected to digital pin 2 of the Arduino to provide pulse signal data, ensuring accurate measurement of the dispensed water volume. This real-time data is sent to the Arduino Uno, allowing the system to precisely calculate and control the total amount of water poured, ensuring it matches the user's selection (50 ml, 100 ml, or 200 ml). The accuracy of the YF-S401 sensor, which operates on the hall-effect principle, is crucial for meeting the user's exact requirements, preventing water wastage or shortages that could affect applications like plant irrigation or livestock watering. Furthermore, the ability to monitor water flow in real time opens up potential applications in scenarios requiring high precision, such as dosing water for hydroponic systems or maintaining consistent water levels in small-scale aquaponics setups. The relay module, interfaced with the Arduino via pin 8 (IN), is responsible for controlling the operation of the mini submersible pump, a high-power device in the system. When the Arduino receives a command from the user (via the RemoteXY app or push buttons), it sends a signal to the relay to activate the pump, initiating the water pouring process. Once the YF-S401 sensor confirms that the dispensed water volume has reached the desired amount, the Arduino sends a signal to the relay to cut off power, stopping the pump immediately. The use of a relay provides two primary benefits:

safety and stability. It isolates the low-power control circuit (Arduino) from the high-power circuit (pump), reducing the risk of damage due to overload or short circuits, while also prolonging the system's lifespan by ensuring the pump operates only when necessary. This precise control mechanism, combined with the dual control options and real-time monitoring, makes the Smart Water Pouring System a robust and versatile solution for automated water management.

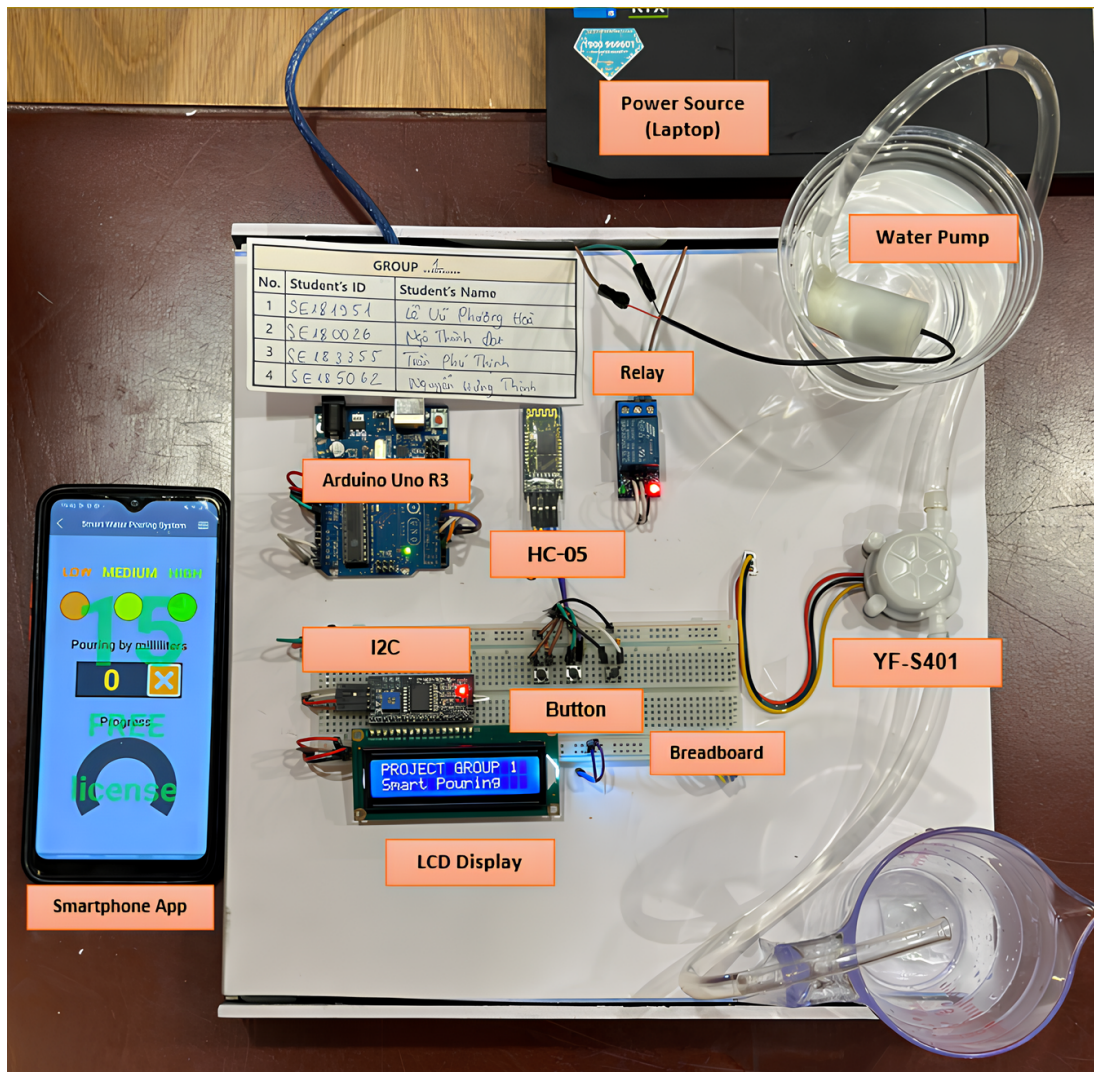


Fig. 12. Smart Water Pouring System Overview.

B. Experimental Results

The Smart Water Pouring System operates by dispensing water based on user-selected predefined levels or custom inputs, ensuring precise and efficient water management. The process begins when the user selects one of the predefined water levels—"Low" (50 ml), "Medium" (100 ml), or "High" (200 ml)—using either the physical push buttons (connected to pins A4, A5, A6 on the Arduino) or the corresponding buttons on the RemoteXY app interface. Alternatively, the user can enter a custom water volume in milliliters (ml) via the text input field on the RemoteXY app. Once the command is initiated—either by pressing a button or submitting a custom value—the Arduino Uno receives the command via the HC-05 Bluetooth module (connected to pins 10 and 11) if using the app, or directly through the physical buttons if using local control. The Arduino then prepares the system by activating the necessary components, including the mini submersible pump (controlled via a relay on pin 8) and the YF-S401 water flow sensor (connected to pin 2 for pulse signal data), to begin the water dispensing process.

When the system starts, the Arduino sends a signal to the relay to activate the pump, initiating water flow to the designated area, such as a plant bed or container. The YF-S401 flow sensor continuously measures the volume of water passing through, using its hall-effect mechanism to generate pulse signals that the Arduino converts into a real-time water volume reading. This data is compared against the target volume (50 ml, 100 ml, 200 ml, or the custom value entered via RemoteXY). If the sensor

detects any sudden changes in flow—such as blockages or variations in pump performance—the Arduino adjusts the pump’s operation by recalibrating the dispensing time to ensure accuracy. Once the YF-S401 sensor confirms that the dispensed volume matches the target, the Arduino sends a signal to the relay to deactivate the pump, immediately stopping the water flow. This automatic shut-off mechanism prevents over-dispensing, promoting water conservation and efficiency.

Throughout the process, the system provides real-time feedback to the user. The 16×2 LCD display (connected via I2C on pins A4 and A5) shows critical information, including the current flow rate, the target volume, and the cumulative amount of water dispensed, allowing users to monitor the system’s progress locally. Simultaneously, if the RemoteXY app is in use, it updates the display elements to reflect the same data, such as the current water level and the total dispensed volume. Upon completion of the watering cycle, the app sends a notification confirming the exact amount of water dispensed (e.g., ”Dispensed: 50 ml”), ensuring transparency and allowing the user to verify that the system has met the specified requirements. The system then enters standby mode, awaiting further commands from the user via the app or physical buttons. In this mode, the Arduino remains active, ready to respond quickly to new tasks while minimizing power consumption by keeping the pump and other components off until needed. This precise, automated process—combined with dual control options and real-time feedback—ensures a user-friendly experience, making the Smart Water Pouring System an efficient solution for applications like plant irrigation, where accurate water dosing is critical.



Fig. 13. Water testing process in Smart Water Pouring System via RemoteXY interface.

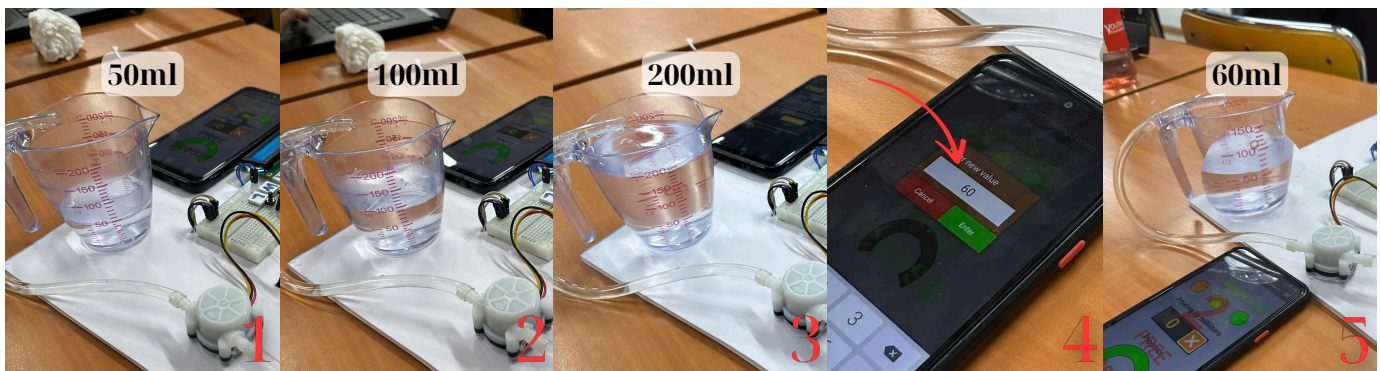


Fig. 14. Result of after testing in Smart Water Pouring System via RemoteXY interface.

C. Discussion

The Smart Water Pouring System, developed with an Arduino Uno R3, mini submersible pump, YF-S401 flow sensor, 16×2 LCD display, push buttons, and HC-05 Bluetooth module, showcases a balance of innovative features and practical challenges that warrant further discussion. One of its standout advantages is its automation capability, driven by the YF-S401 flow sensor's precise measurement of water volumes (50 ml, 100 ml, 200 ml, or custom inputs via the RemoteXY app). This precision ensures optimal water usage, which is particularly beneficial for small-scale applications like plant irrigation or hydroponics, where overwatering or underwatering can harm plant health. The system's ability to reduce manual labor through automation is a significant benefit, especially for users such as hobbyists or small-scale farmers who seek efficient, hands-free solutions. Additionally, the integration of the HC-05 Bluetooth module enables wireless control within a 10-meter range, providing flexibility and convenience through the RemoteXY app. This feature, combined with real-time monitoring via the LCD and app, enhances user interaction and makes the system an excellent educational tool for students learning about IoT, sensor integration, and embedded systems. The dual control options—wireless via Bluetooth and local via push buttons—further increase its accessibility, catering to both tech-savvy users and those who prefer simplicity.

However, the system's limitations reveal several challenges that impact its broader applicability. Its reliance on a laptop or USB power source restricts mobility, making it unsuitable for outdoor or off-grid scenarios without modification. This dependency could be mitigated by adopting a rechargeable battery or solar panel, but such additions would increase costs and complexity, a trade-off that must be considered for practical deployment. The use of a breadboard and jumper wires, while ideal for prototyping, introduces wiring complexity and a higher risk of connection failures over time, which could compromise reliability in long-term use. Transitioning to a custom PCB would improve stability but requires additional design effort and expense, potentially limiting its feasibility for educational projects where cost and simplicity are priorities. The HC-05 Bluetooth module's 10-meter range, while sufficient for small setups, is a significant constraint for large-scale irrigation applications, such as in expansive gardens or farms. Adding a Wi-Fi module (e.g., ESP8266) could extend the range and enable cloud connectivity, but this would increase power consumption and system complexity, potentially negating Bluetooth's low-power advantage.

Durability is another concern, as the prototype lacks weatherproofing, making it vulnerable to outdoor conditions like rain, dust, or extreme temperatures. A weatherproof enclosure (e.g., IP65-rated) could address this, but it would add to the system's cost and weight, which may not align with its intended use as a lightweight educational tool. The RemoteXY app, while user-friendly, lacks advanced features like scheduling, push notifications, or data logging, limiting its utility for precision agriculture or long-term monitoring. Enhancing the app would improve functionality but could make it less intuitive for beginners, a key audience for this system. The YF-S401 flow sensor's accuracy, while generally reliable, can be affected by debris or water pressure variations, leading to potential inconsistencies. Regular maintenance and calibration can mitigate this, but these requirements may deter users seeking a low-maintenance solution. Finally, the absence of error-handling mechanisms means that malfunctions, such as sensor failures or pump issues, may go undetected, reducing reliability. Implementing error detection in the firmware would enhance robustness but could increase development time and complexity, a consideration for educational projects with limited scope.

IV. CONCLUSION

The Smart Water Pouring System prototype, built with an Arduino Uno R3, mini submersible pump, YF-S401 flow sensor, 16×2 LCD, push buttons, and HC-05 Bluetooth module, demonstrates significant potential as an automated watering solution. Its strengths, including precise water dispensing, wireless control via the RemoteXY app, and real-time monitoring, make it an efficient tool for small-scale irrigation and an excellent educational platform for learning about IoT and embedded systems. However, limitations such as power dependency, limited Bluetooth range, lack of weatherproofing, and the absence of advanced app features and error handling restrict its practical deployment. Addressing these challenges—through solutions like a battery power source, Wi-Fi integration, a weatherproof enclosure, app enhancements, and firmware improvements—can enhance its reliability, scalability, and applicability, making it a more robust solution for both educational and real-world use.

TABLE III
AUTHOR'S CONTRIBUTION

#	Student ID	Student Name	Tasks	Contribution
1	SE181951	Le Vu Phuong Hoa	Task division, Content checking, Program Arduino, and support	25%
2	SE180026	Ngo Thanh Dat	Draw Circuit Schematic, Prepare presentation, design slides, videos and support	25%
3	SE183355	Tran Phu Thinh	Draw block diagram, flowchart, Programming Flowchart, and support	25%
4	SE185062	Nguyen Hung Thinh	Drawcharts, Prepare presentation, find material and support	25%
Total				100%

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